

SKILLSHEET

Finding simple interest

Simple interest is calculated using the formula

$$I = \frac{P \times r \times T}{100}$$

where I is the simple interest (\$)
 P is the principal (the amount borrowed or invested) (\$)
 r is the rate of interest per annum (% p.a.)
 T is the time period for which the money is borrowed or invested.

WORKED EXAMPLE

Calculate the simple interest earned on an investment of \$5000 at 6.5% p.a. for 3.5 years.

THINK

- Write the simple interest formula and state the values of P , r and T .
- Substitute the values into the formula to find the value of I .
- Write a statement.

WRITE

$$I = \frac{P \times r \times T}{100}$$

where $P = \$5000$, $r = 6.5\%$ and $T = 3.5$ years

$$= \frac{5000 \times 6.5 \times 3.5}{100}$$

$$= 1137.5$$

The simple interest earned is \$1137.50.

Try these

Calculate the amount of simple interest on the following loans or investments.

- 1 \$4500 is invested at 3.65% p.a. for 12 months.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{ and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

- 2 \$7800 is invested at 4.95% p.a. for 18 months.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{ and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

- 3 \$4500 is borrowed at 8.25% p.a. for 2 years.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{ and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

- 4 \$14 500 is invested at 5.25% p.a. for 15 months.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{ and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

(continued)

SKILLSHEET

(continued)

- 5** \$25 000 is borrowed at 8.95% p.a. for 5 years.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

- 6** \$9870 is invested at 4.75% p.a. for 2 years.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

- 7** \$9000 is invested at 5.68% p.a. for 4.5 years.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

- 8** \$15 000 is borrowed at 7.0% p.a. for 5.5 years.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

- 9** \$12 000 is borrowed at 8.9% p.a. for 3 years.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

- 10** \$45 000 is invested at 4.7% p.a. for 24 months.

$$I = \frac{P \times r \times T}{100} \quad P = \dots\dots\dots r = \dots\dots\dots \text{and } T = \dots\dots\dots \text{ years}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

SKILLSHEET — ANSWERS

SKILLSHEET

Finding simple interest

- | | |
|---------------|------------|
| 1 \$164.24 | 2 \$579.15 |
| 3 \$742.50 | 4 \$951.56 |
| 5 \$11 187.50 | 6 \$937.65 |
| 7 \$2300.40 | 8 \$5775 |
| 9 \$3204 | 10 \$4230 |